Reg. No. | | | | | | | | | | | [14CS-IS45]

FOURTH SEMESTER B.Tech UNIVERSITY EXAMINATION, JUNE–2017

THEORY OF COMPUTATION

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed.

PART – A

1. a. Define with an example for each: **03**

i. language

ii. string

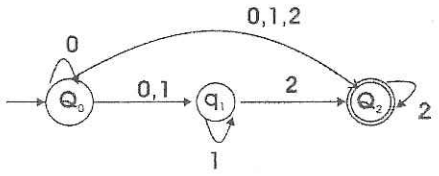
iii. alphabet

b. Construct DFA for the following languages: **06**

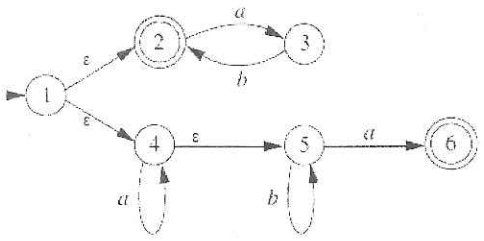
i. Strings of a's and b's containing either ab or bba

ii. Strings of 0's and 1's that either start with 0 and has odd length OR start with 1 and has even length

c. Convert the following NFA into an equivalent DFA. **05**



2. a. Convert the following ε –NFA into an equivalent DFA. **07**



b. Differentiate the extended transition functions for DFA and ε –NFA with examples. **04**

C. Write a note on the applications of finite automata. **03**

3. a. Write regular expressions for the following languages: **08**

i. $L= \{a^{2n}b^{2m} \mid n \geq 0, m \geq 0\}$

ii. Strings that do not end with 01 for $\Sigma=\{0,1\}$

iii. $L= \{x \in \{0,1\}^* \mid x$ ends with 1 and does not contain 00$\}$

iv. Strings of even length over $\Sigma=\{0,1\}$

b. Obtain a RE for the DFA below by eliminating states **06**

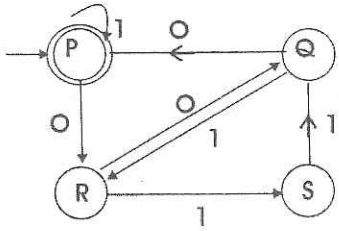


Fig.3b

4. a. State and prove the pumping lemma for regular languages. **06**

b. Prove that the language L = { a^p : p is a prime} is not a regular language **03**

C. Minimize the following DFA. **05**

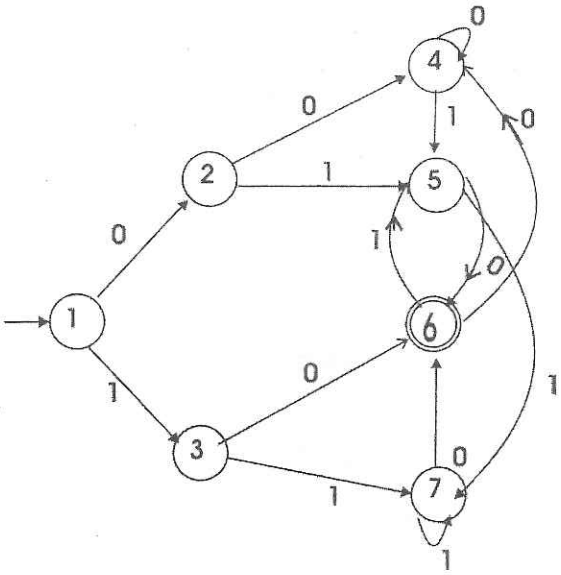


Fig.4c

PART B

5. a. Write CFG for the following languages: **08**

i. L={ $0^i1^j2^k$ | j > i+k}

ii. L ={ w | w ε {a,b}*, $n_a(w) \neq n_b(w)$ }

iii. L = { $a^i b^j$ |i ≤ 2j}

iv. The set of even length strings in {a,b}* with the two middle symbols equal.

b. Define ambiguous grammars. Is the following grammar ambiguous? **04**

S → iCtS | iCtSeS | a

C → b

c. Obtain the RMD for the string "aabbab" given the grammar **02**

S→ aB | bA

A → bAA | aS | a

B → aBB | bS | b

6. a. Design a PDA to accept the language $\{ 0^m1^n0^m \mid m,n \geq 1|$. Trace the moves of the PDA on the string 00100. **08**

b. Explain how a PDA can be constructed to accept the language of a CFG. Find a PDA to accept the language of the grammar $S \rightarrow aA$, $A \rightarrow aA \mid bA \mid a \mid b$ **06**

7. a. Convert the following grammar into Chomsky Normal Form. **08**

$S \rightarrow AACD$, $A \rightarrow aAb \mid \varepsilon$, $C \rightarrow aC \mid a$, $D \rightarrow aDa \mid bDb \mid \varepsilon$

b. Prove that if L ia CFL and R is a regular language, then $L \cap R$ is also a CFL. **03**

c. Use the pumping lemma for CFL to prove that the language $L=\{0^n1^n2^n \mid n \geq 1 \}$ is not context-free. **03**

8. a. Define Turing machine. Explain it's working with a neat diagram **05**

b. Design a turing machine to accept the language $L= \{a^nb^{2n} \mid n \geq 1\}$. Give the computation for the input aabbbb **09**